Theoretical Foundation

Mixed-Integer Linear Programming (MILP) Framework

A mixed-integer linear program is an optimisation problem where all constraints and the objective function are linear, but only a subset of the decision variables may have continuous values. A general MILP is written as shown in Equations (26) and (27) [1]:

$$\min_{x,y} c^T x + h^T y \tag{26}$$

$$s.t.\, Ax + Gy \leq b \begin{cases} x \geq 0, & x\epsilon\mathbb{R}^n \\ y \geq 0, & y\epsilon\mathbb{Z}^p \end{cases} \tag{27}$$

where $x\epsilon\mathbb{R}^n$ is continuous decision variable; $y\epsilon\mathbb{Z}^p$ is integer decision variable (if y is restricted to $\{0,1\}^p$ then it's a mixed-binary program); c, h are cost coefficient vectors; A, G are constraint coefficient vectors; and b is a right-hand-side vector [62].

MILP Formulation and Optimization

System Description and Optimization Framework

MILP framework is developed to determine the optimal annual operation and H_2 -storage capacity of the PV-powered electrolyzer system. The optimization is performed on an hourly basis over a one-year period, with N = 8760 time periods.

The hourly PV-driven electrolyzer operation is obtained from the Simulink model. Therefore, the optimization model does not re-optimize the PV-electrolyzer conversion process. The optimization uses hourly PV contribution obtained from the simulation as operational input and determines the additional grid-electricity allocation and storage requirements.

Input Data and Assumptions

The principal time-dependent inputs are:

- Available PV electricity;
- PV electricity supplied to the electrolyzer;
- H_2 production associated with PV operation;
- Industrial electrical demand;
- Hourly H_2 demand;
- Electricity purchase price; and
- PV feed-in tariff.

The electrolyzer has a rated electrical power of $P_{Elec}^{max} = 10\,\text{kW}$. A minimum operating load of 10 % of the rated electrolyzer power is imposed when the electrolyzer is operating: $P_{Elec,}^{min} = 0.10\,P_{Elec}^{max} = 1\,\text{kW}$.

Electrolyzer is not permitted to operate during predefined shutdown periods. In the considered roastery operating schedule, Sundays are excluded from operation.

Decision Variables

The principal continuous decision variables considered in the optimization model are defined for every hour t and summarized in Table 1. Continuous variables are allowed to take any real-values quantity within the bounds imposed by the optimization constraints. Binary variable can only take the values 0 or 1. For the high pressure (HP) storage, the additional variables mentioned in

Table 2 are introduced.

Table 1: Continuous decision variables used in the MILP formulation.

Variable	Description	Unit	Type
$E_{Elec,t}^{Grid}$	Grid electricity supplied to the electrolyzer	kWh	Continuous
H_t	H_2 produced during hour t	kg	Continuous
S_t	H_2 storage level	kg	Continuous
C_{H_2}	H_2 storage capacity	kg	Continuous
$E_{Load,t}^{PV}$	PV electricity supplied to the auxiliary/site electrical load	kWh	Continuous
$E_{Load,t}^{Grid}$	Grid electricity supplied to the auxiliary/site electrical load	kWh	Continuous
u_t	Electrolyzer operating state	-	Binary

Table 2: Continuous decision variables specific to HP storage configuration.

Variable	Description	Unit	Type
E_t^{Comp}	Compressor electricity consumption	kWh	Continuous
$E_{Comp,t}^{PV}$	PV electricity supplied to compressor	kWh	Continuous
$E_{Comp,t}^{Grid}$	Grid electricity supplied to compressor	kWh	Continuous

Objective Function

The objective of optimization is to minimize the annual cost associated with grid-electricity consumption and H_2-storage capacity, while accounting for revenue from exported surplus PV electricity. The objective function can be expressed by Equation (33) –

$$\min Z = C_{Grid} + C_{H_2,stor} - R_{PV} \tag{33}$$

where C_{Grid} represents the annual cost of electricity purchased from the grid (see Equation (34–35)), $C_{H_2,stor}$ is the annualized H_2-storage cost (see Equation (36)), and R_{PV} represents revenue obtained from exporting the surplus PV electricity (see Equation (37)).

For the low-pressure configuration –

$$C_{Grid} = \sum_{t=1}^{T} p_t \left(E_{Elec,t}^{Grid} + E_{Load,t}^{Grid}\right) \tag{34}$$

where p_t is the electricity price in hour t.

For the high-pressure configuration, compressor electricity consumption is additionally included –

$$C_{Grid} = \sum_{t=1}^{T} p_t \left(E_{Elec,t}^{Grid} + E_{Load,t}^{Grid} + E_{Comp,t}^{Grid}\right) \tag{35}$$

$$C_{H_2,stor} = c_{H_2,stor}^{ann} C_{H_2} \tag{36}$$

where $c_{H_2,stor}^{ann}$ is the annualized storage cost per unit storage capacity.

$$R_{PV} = \sum_{t=1}^{T} FIT_t \left(E_{exp,t}^{PV}\right) \tag{37}$$

where FIT_t is the applicable feed-in tariff. As a result, the optimizer seeks the economically optimal combination of grid electricity and H_2-storage capacity while satisfying all operational constraints.

Electrolyzer Operating Constraints

The total electricity supplied to the electrolyzer is defined by Equation (38) –

$$E_{Elec,t} = E_{Elec,t}^{PV} + E_{Elec,t}^{Grid}. \tag{38}$$

The electrolyzer operating state is represented by the binary variable $u_t \epsilon \{0,1\}$ as ON/OFF state. Its maximum operating limit is therefore $E_{Elec,t} \leq P_{Elec}^{max} u_t$.

Equations (39) and (40) ensure that the electrolyzer can only operate in one of two states:

$$u_t = 0 \rightarrow E_{Elec,t} = 0, \tag{39}$$

or

$$u_t = 1 \rightarrow P_{Elec}^{min} \leq E_{Elec,t} \leq P_{Elec}^{max}. \tag{40}$$

The operating state is additionally restricted by the predefined availability schedule: $u_t \leq A_t$, where $A_t = 0$ during scheduled shutdown periods and $A_t = 1$ during permitted operating periods. Grid contributes to satisfying the system requirements and is bounded according to the electrolyzer capacity: $0 \leq E_{Elec,t}^{Grid} \leq P_{Elec}^{max} - E_{Elec,t}^{PV}$.

Site-load Balance

The site load is represented by L_t. The total site loads for the system are assumed and divided into three parts: base electricity load of the industry, auxiliary load (electrical consumption associated with the roasting process, ventilation, control systems), and roaster electrical load (electricity for ignition, PLC control, combustion air fans). The base load is 5 kWh/day from Monday to Saturday, auxiliary load is 10 kWh/day during roasting hours, and roaster electrical load is 10 kW on roasting days. The total site load can be supplied by either PV electricity or the grid electricity: $E_{Load,t}^{PV} + E_{Load,t}^{Grid} = L_t$.

The corresponding bounds are:

$$0 \leq E_{Load,t}^{PV} \leq L_t$$

and

$$E_{Load,t}^{Grid} \geq 0.$$

Hydrogen-storage Balance

The H_2 storage system is represented using an hourly material balance as shown in Equations (41) and (42).

$$S_1 = S_0 + H_t - D_t, \qquad t = 0 \tag{41}$$

$$S_1 = S_{t-a} + H_t - D_t, \qquad t > 1. \tag{42}$$

Here, D_t denotes the hourly H_2 demand of the roasting process. The storage inventory is bounded by $0 \leq D_t \leq C_{H_2}$ where C_{H_2} is itself an optimization variable. Hence, the optimized storage capacity implies the maximum H_2 inventory required during the optimization.

Case 1: Low-pressure H_2 Storage

In case 1, H_2 is stored directly in the low-pressure buffer storage after electrolyzer production. No additional compression electricity requirement is therefore included. The storage formulation is

subject to a cyclic annual condition $S_T = S_0$. This cyclic boundary means that the storage tank must finish the simulated year with the same amount of H_2 with which it started. The principal storage-sizing variable is therefore $C_{H_2}^{LP}$, which is the required low-pressure H_2 storage capacity.

Case 2: High-pressure H_2 Storage

Case 2 extends the LP-storage formulation by introducing H_2 compression prior to storage. The compressor electricity consumption is calculated from the produced H_2: $E_t^{Comp} = SEC_{Comp} H_t$, where SEC_{Comp} is the specific energy consumption by compressor and obtained from the Simulink model B. The compressor electricity can be supplied by PV or the grid as shown by Equation (43) –

$$E_t^{Comp} = E_{Comp,t}^{PV} + E_{Comp,t}^{Grid}. \quad (43)$$

As a result, there is additional electricity demand in the system, and it creates additional competition for PV electricity. The HP storage capacity is represented by $C_{H_2}^{HP}$, with $0 \leq S_t^{HP} \leq C_{H_2}^{HP}$. Therefore, case 2 introduces three additional hourly decision variables, E_t^{Comp}, $E_{Comp,t}^{PV}$, $E_{Comp,t}^{Grid}$, and replaces the LP-storage representation. The annualized HP-storage cost is calculated from the optimized capacity as $C_{HP}^{ann} = c_{HP}^{ann} C_{H_2}^{HP}$.

PV Energy Allocation

The available PV electricity in each hour is allocated to competing system demands. The general PV balance is represented by Equation (44) –

$$E_{available,t}^{PV} = E_{Elec,t}^{PV} + E_{Load,t}^{PV} + E_{Comp,t}^{PV} + E_{exp,t}^{PV} + E_{curt,t}^{Grid}. \quad (44)$$

Solution Procedure

The formulated problem is solved using *intlinprog* – optimization algorithm of MATLAB – by determining globally optimal solution within the specified optimality tolerance as shown in Appendix 10.3 and 10.4. The resulting solution provides:

- annual PV electricity consumed by the electrolyzer,
- annual grid electricity supplied to the electrolyzer,
- annual H_2 production,
- annual grid electricity consumption,
- hourly H_2-storage path,
- required H_2-storage capacity,
- compressor electricity consumption for case 2;
- PV electricity exported to the grid;
- annualized storage cost; and
- total annual system cost.

Reference

[1] J. P. Vielma, ‘Mixed Integer Linear Programming Formulation Techniques’, *SIAM Rev.*, vol. 57, no. 1, pp. 3–57, Jan. 2015, doi: 10.1137/130915303.